

Multiple Choice (10 marks)

1. Which of the following sets has the greatest cardinality?
 - (a) The set of all finite subsets of $\mathbb{R}$
 - (b) $\mathbb{R}$
 - (c) The set of all infinite subsets of $\mathbb{R}$
 - (d) The set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$
 - (e) The set of all functions from $\{0, 1\}$ to $\mathbb{R}$
2. There are 25 suitcases, 5 of which are damaged. Three suitcases are selected at random. What is the probability that exactly 2 are damaged?
 - (a) $2/50$
 - (b) $1/25$
 - (c) $1/30$
 - (d) $3/75$
 - (e) $4/100$
3. A grocery store sells a bag of 5 lemons for \$2.00. What is the unit cost of each lemon in the bag?
 - (a) \$1.50
 - (b) \$0.50
 - (c) \$0.20
 - (d) \$0.60
 - (e) \$0.10
4. What number makes the equation $35 / ? = 7$ true?
 - (a) 5
 - (b) 25
 - (c) 7
 - (d) 6
 - (e) 10
5. Compute the product in the given ring. $(12)(16)$ in $\mathbb{Z}_{24}$
 - (a) 24
 - (b) 28

- (c) 20
- (d) 16
- (e) 8

True / False (10 marks)

Answer True or False

6. True or False: The cost of one pencil is 42 cents, assuming the price relationship between pencils and pens remains consistent.
7. True or False: The solution to the system of equations using Cramer's rule is given by the vector $[-1, 3, 1]$.
8. True or False: There are 240 ways to divide a set of 5 elements into 2 non-empty ordered subsets.
9. True or False: The expression $1990 \times 1991 - 1989 \times 1990$ evaluates to the counting number 3970.
10. True or False: The solution to the differential equation $dy/dt =$ with the initial condition $y(1) = 1$ is $y(4) = 7.789$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the division of the expression 2,314 by 4. Discuss the process of obtaining the quotient and remainder, and explain the significance of these results in mathematical terms.
12. Determine the smallest positive integer that satisfies the conditions of leaving specific remainders when divided by 8, 3, and 11. Discuss your solution process and reasoning.

End of Paper